

Random events and their probabilities

Count the outcomes, count the favourable ones, and divide — but only when they are equally likely.

LESSON 79–80

2 × 40 MIN

ALGEBRA 11, §29

IGX 24.2

LESSON CLOCK

15'

20'

20'

20'

5'

The sample space	15 min
The classical definition	20 min
Counting with combinations	20 min
Relative frequency	20 min
Homework	5 min

LEARNING OBJECTIVES

- Describe a trial by its sample space.
- Compute a probability from equally likely outcomes.
- Estimate a probability from relative frequency, and say when that is required.
- Use combinatorics to count outcomes that cannot be listed.

TEXTBOOK

National	pp. 342–352
Cambridge	Core & Extended, pp. 629–634
Workbook	Exercise 24.2

01

Explanation

The sample space

Grade 11 begins where Grade 10 left off, but the counting is harder: the sample spaces are now too big to list, and the combinatorics of Quarter III does the listing instead.

$$P(A) = \frac{m}{n}, n = |U|, m \text{ the favourable count}$$

TRIAL	$ U $	COUNTED BY
two dice	36	6^2
a five-card hand from 52	2 598 960	$C(52, 5)$
an arrangement of 5 books	120	$5!$
a three-digit code, digits repeatable	1000	10^3
a committee of 3 from 10	120	$C(10, 3)$

ORDER, OR NO ORDER?

A **hand** of cards, a **committee**, a **selection** — order does not matter, use $C(n, k)$. A **code**, an **arrangement**, a **podium** — order matters, use $P(n, k)$ or a power. Choosing the wrong one changes n by a factor of $k!$.

The classical definition in use

The pattern is always the same: count n , count m , divide — and use the **same counting method** for both.

EVENT	M	N	P
two dice total 7	6	36	$\frac{1}{6}$
a committee of 3 from 6 men and 4 women is all women	$C(4,3) = 4$	$C(10,3) = 120$	$\frac{1}{30}$
two aces in a five-card hand	$C(4,2) \cdot C(48,3)$	$C(52,5)$	≈ 0.0399
a random arrangement of $ABCDE$ starts with A	$4! = 24$	$5! = 120$	$\frac{1}{5}$

COUNT BOTH WITH THE SAME METHOD

If n is a number of **unordered** selections, m must be too. Mixing $C(52, 5)$ below with an ordered count above gives an answer wrong by $5!$ — a factor of 120.

Counting with combinations

$$C(n, k) = \frac{n!}{k!(n-k)!}, P(n, k) = \frac{n!}{(n-k)!}$$

The standard shape of a selection problem: choose some from one group and the rest from another, then **multiply**.

Example. A committee of 4 from 7 men and 5 women, with exactly 2 women:

$$\frac{C(5, 2) \cdot C(7, 2)}{C(12, 4)} = \frac{10 \times 21}{495} = \frac{210}{495} = \frac{14}{33}$$

“AT LEAST” MEANS ADD THE CASES – OR USE THE COMPLEMENT

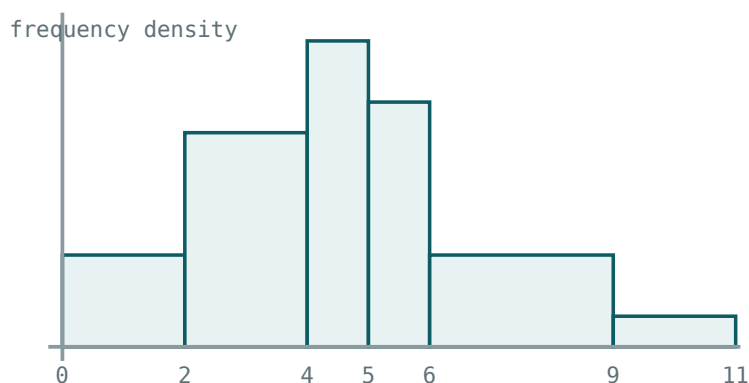
“At least one woman” on that committee: either add the cases with 1, 2, 3, 4 women, or compute $1 -$

$$\frac{C(7, 4)}{C(12, 4)} = 1 - \frac{35}{495} = \frac{92}{99}. \text{ The complement is one line instead of four.}$$

Relative frequency

When outcomes are not equally likely, probability must be measured:

$$P(A) \approx \frac{\text{times A occurred}}{\text{number of trials}}$$



with unequal widths the AREA is the frequency

Frequencies from an experiment — the estimate that stands in for a count.

The estimate steadies as the number of trials grows. That is the **law of large numbers**, and it is the reason a table of experimental data can stand in for a theoretical probability at all.

SITUATION	WHICH DEFINITION
a fair die	classical — symmetry guarantees equal outcomes
a drawing pin	statistical — no symmetry to appeal to
a component failing in a year	statistical — from records
a point in a region	geometric — a ratio of measures

Expected number. In N trials with probability p , expect Np successes. In 500 throws of a fair die, expect ≈ 83 sixes — the number the binomial distribution of Lessons 86–89 will make precise.

02

Worked examples

EXAMPLE 1 A committee of 3 is chosen from 6 men and 4 women. Find the probability that it is all women.

1

$$n = C(10, 3) = 120$$

2

$$m = C(4, 3) = 4$$

3

$$P = \frac{4}{120} = \frac{1}{30}$$

ANSWER

$$\frac{1}{30} \approx 0.033$$

EXAMPLE 2 A committee of 4 is chosen from 7 men and 5 women. Find the probability of exactly 2 women.

$$(1) \quad n = C(12, 4) = 495$$

$$(2) \quad m = C(5, 2) \cdot C(7, 2) = 10 \times 21 = 210$$

Choose from each group, then multiply.

$$(3) \quad P = \frac{210}{495} = \frac{14}{33}$$

ANSWER

$$\frac{14}{33} \approx 0.424$$

EXAMPLE 3 The letters of *MATHS* are arranged at random. Find the probability that the arrangement begins with a vowel.

$$(1) \quad n = 5! = 120$$

$$(2) \quad \text{Only A is a vowel; the rest arrange in } 4! = 24 \text{ ways.}$$

$$(3) \quad P = \frac{24}{120} = \frac{1}{5}$$

ANSWER

$$\frac{1}{5}$$

EXAMPLE 4 A die is thrown 600 times and shows a six 118 times. Compare with the theoretical value.

① Relative frequency $\frac{118}{600} \approx 0.1967$

② Theoretical $\frac{1}{6} \approx 0.1667$

③ Expected number $600 \times \frac{1}{6} = 100$

The excess of 18 is within ordinary variation.

ANSWER

0.197 against 0.167; expected 100

03 Interactive model

Bring a pack of cards and deal ten hands of five; count the aces and compare with the computed probability.

Counting the sample space

A hand of 5 cards from 52 is counted by:

$$52^5$$

$$C(52, 5)$$

$$P(52, 5)$$

$$5!$$

A 3-digit code with repeats:

$$C(10, 3)$$

$$10^3$$

$$P(10, 3)$$

$$3!$$

Arrangements of 5 books:

$$5$$

$$25$$

$$120$$

$$32$$

A committee of 3 from 10:

120

720

1000

30

Two dice give:

12

21

36

6

For a bent drawing pin use:

the classical definition

relative frequency

geometry

nothing

Expected sixes in 600 throws:

60

100

120

600

“At least one” is quickest by:

listing

the complement

a tree

a table

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Random event	Tasodifiy hodisa	Случайное событие
Sample space	Elementar hodisalar fazosi	Пространство исходов
Equally likely	Teng imkoniyatli	Равновозможные
Favourable outcome	Qulay natija	Благоприятный исход
Relative frequency	Nisbiy chastota	Относительная частота
Combination	Kombinatsiya	Сочетание
Permutation	O‘rin almashtirish	Перестановка
Expected number	Kutilayotgan son	Ожидаемое число
Certain event	Muqarrar hodisa	Достоверное событие
Impossible event	Mumkin bo‘lmagan hodisa	Невозможное событие

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The classical definition needs outcomes that are:

A selection where order matters is counted by:

m and n must be counted:

The law of large numbers says the relative frequency:

grows

settles towards a fixed value

oscillates for ever

reaches 1

Choosing 2 from one group and 2 from another needs:

addition

multiplication

subtraction

division

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $C(10, 3)$

ANSWER 120

2. $C(52, 2)$

ANSWER 1326

3. $5!$

ANSWER 120

4. $P(6, 2)$

ANSWER 30

5. Outcomes for two dice

ANSWER 36

6. P a die shows 4

ANSWER $\frac{1}{6}$

7. Expected sixes in 600 throws

ANSWER 100

Medium · the standard question

7 PROBLEMS

1. P a committee of 3 from 6 men and 4 women is all women

ANSWER $\frac{1}{30}$

2. P exactly 2 women on a committee of 4 from 7 men and 5 women

ANSWER $\frac{14}{33}$

3. P the arrangement of *MATHS* begins with A

ANSWER $\frac{1}{5}$

4. P two dice total 9

ANSWER $\frac{1}{9}$

5. P a five-card hand is all hearts

ANSWER $\frac{C(13,5)}{C(52,5)} \approx 0.000495$

6. P at least one woman on a committee of 4 from 7 men and 5 women

ANSWER $\frac{92}{99}$

7. Relative frequency of 118 sixes in 600 throws

ANSWER ≈ 0.197

Hard · stretch and reason

7 PROBLEMS

1. P a five-card hand contains exactly two aces

ANSWER ≈ 0.0399

2. P a five-card hand is a full house

ANSWER ≈ 0.00144

3. P the letters of *LEVEL* arrange to give *LEVEL*

ANSWER $\frac{1}{30}$

4. From 10 people, P two named people sit together in a row

ANSWER $\frac{1}{5}$

5. From 10 people in a circle, P two named people sit together

ANSWER $\frac{2}{9}$

6. P a random 4-digit PIN has all different digits

ANSWER $\frac{P(10,4)}{10^4} = 0.504$

7. A committee of 5 from 12 must contain a given person: P

ANSWER $\frac{5}{12}$

07 Homework

Homework — 5 tasks

Write n and m on separate lines, each with the counting method named.

1. A committee of 3 is chosen from 5 men and 7 women. Find the probability that it contains exactly one man.
2. The letters of *NUMBER* are arranged at random. Find the probability that the arrangement begins and ends with a consonant.

3. Two cards are drawn from 52 without replacement. Find the probability that both are hearts.
4. A die is thrown 300 times and shows an even number 168 times. Find the relative frequency and compare it with the theoretical value.
5. Explain in two sentences why m and n must be counted by the same method.